

POINTE DU RAZ, France

WMO: **071030**Lat: **48.0389N**Lon: **4.7319W**Elev: **89**StdP: **100.26**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.8	2.0	-4.8	2.5	1.9	-3.2	2.9	3.2	23.5	8.6	22.0	8.6	7.0	70	0.804

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	22.5	17.7	20.7	17.1	19.6	16.6	18.7	21.1	17.8	19.6	17.2	18.7	5.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	19.5	17.1	12.3	18.6	16.6	11.9	18.0	53.3	21.2	50.5	19.7	48.8	18.8	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.3	16.2	-1.8	25.7	1.2	2.4	-2.7	27.5	-3.5	28.9	-4.2	30.2	-5.1	31.9		
(5)			-3.4	19.8	1.2	1.5	-4.3	20.9	-5.0	21.8	-5.7	22.6	-6.6	23.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	8.2	7.6	8.8	10.6	12.8	15.1	16.6	16.8	16.0	14.1	11.1	8.7
	DBStd	3.91	2.65	2.55	2.38	1.99	1.93	1.71	1.71	1.33	1.56	1.86	2.12	2.64
	HDD10.0	265	63	72	49	14	1	0	0	0	0	1	13	53
	HDD18.3	2247	313	302	296	233	172	98	60	52	73	133	218	298
	CDD10.0	1077	8	3	12	31	88	154	205	211	179	127	46	13
	CDD18.3	15	0	0	0	0	1	2	6	4	2	0	0	0
	CDH23.3	32	0	0	0	0	2	3	12	11	4	0	0	0
	CDH26.7	2	0	0	0	0	1	0	1	1	0	0	0	0
Wind	WSAvg	7.9	9.6	8.5	8.3	7.5	7.4	6.5	6.4	6.3	6.5	8.3	9.8	10.1
Precipitation	PrecAvg	819	96	79	56	64	53	42	48	50	54	85	100	93
	PrecMax	1091	210	191	164	148	129	91	121	106	144	125	197	166
	PrecMin	589	25	16	8	8	7	9	7	4	4	17	16	15
	PrecStd	112	49	41	34	37	30	24	29	30	34	29	46	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	12.6	17.2	18.3	21.8	23.3	24.8	25.2	23.2	19.1	15.8	13.8
		MCWB	11.7	11.4	11.3	13.4	16.0	18.0	18.0	19.9	17.3	16.8	14.7	13.0
	2%	DB	12.2	11.7	14.4	16.0	18.7	20.5	22.2	21.7	20.6	17.7	15.0	12.9
		MCWB	11.4	10.6	10.5	12.5	14.6	16.6	17.6	18.0	16.9	15.9	13.7	11.9
	5%	DB	11.7	11.2	12.4	14.6	16.9	19.1	20.6	19.9	19.3	16.9	14.4	12.4
		MCWB	10.9	10.2	10.1	11.9	14.1	15.9	17.2	17.2	16.5	15.5	13.3	11.5
	10%	DB	11.2	10.7	11.5	13.2	15.4	17.9	19.3	18.9	18.3	16.3	13.9	12.0
		MCWB	10.4	9.7	9.9	11.2	13.2	15.3	16.6	16.7	16.1	14.9	12.8	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	11.7	12.6	14.2	16.8	18.9	19.2	20.4	19.3	17.4	15.2	13.1
		MCDB	12.4	12.2	15.0	16.3	20.9	21.6	22.5	24.3	21.8	18.3	15.6	13.4
	2%	WB	11.7	11.1	11.5	13.2	15.1	17.2	18.1	18.8	18.0	16.6	14.2	12.4
		MCDB	12.0	11.5	12.8	14.9	17.9	19.5	20.9	20.9	19.4	17.3	14.7	12.7
	5%	WB	11.2	10.6	10.9	12.2	14.3	16.2	17.5	17.8	17.3	16.0	13.6	11.9
		MCDB	11.5	11.0	11.8	13.8	16.4	18.2	19.7	19.1	18.3	16.7	14.2	12.2
	10%	WB	10.6	10.0	10.3	11.5	13.5	15.6	16.8	17.2	16.7	15.5	13.0	11.3
		MCDB	11.0	10.6	11.2	12.9	15.0	17.3	18.6	18.4	17.7	16.2	13.7	11.8
Mean Daily Temperature Range	5% DB	MDBR	2.8	3.1	3.9	4.2	4.1	4.2	4.2	4.1	4.1	3.2	2.9	3.1
		MCDBR	2.2	2.9	5.2	5.7	6.2	6.4	6.3	5.5	5.6	3.5	3.0	2.6
		MCWBR	2.7	2.7	3.7	3.9	3.8	4.0	3.6	3.3	3.6	2.9	3.2	3.3
	5% WB	MCDBR	2.1	2.2	3.6	4.6	5.4	5.4	5.4	4.9	4.5	3.1	2.6	2.5
		MCWBR	2.8	2.5	2.9	3.4	3.6	3.8	3.4	3.2	3.3	2.9	3.2	3.4
Clear-Sky Solar Irradiance	taub	0.337	0.350	0.374	0.381	0.384	0.379	0.380	0.369	0.359	0.358	0.339	0.330	
	taud	2.405	2.395	2.351	2.353	2.391	2.430	2.432	2.480	2.489	2.471	2.431	2.375	
	Ebn at Noon	726	803	839	870	879	884	877	873	849	785	717	679	
	Edh at Noon	69	86	104	114	114	111	109	100	90	79	67	63	
All-Sky Solar Radiation	RadAvg	1.00	1.83	3.06	4.77	5.62	6.08	5.81	4.99	3.99	2.34	1.25	0.84	
	RadStd	0.07	0.17	0.32	0.38	0.48	0.64	0.29	0.39	0.28	0.17	0.09	0.09	